

MOVING CHARGES AND MAGNETISM

✚ Important formulas and concepts directly from the NCERT-

- Magnetic field can be produced due to moving charge and current carrying conductor.

❖ Force on a charge 'q' placed in external field-

$$F = qvB \sin\theta$$

θ = angle b/w 'v' and 'B'

- Direction of force can be given by 'flemming left hand rule'.

❖ When particle thrown with 'v' inside a transverse uniform magnetic field-

$$F = qvB \sin(90^\circ)$$

$$F = qvB$$

- Particle will perform 'uniform circular motion'.

- Radius of circular motion,

$$R = mv/qB = P/qB = (2mK.E.)^{1/2}/qB = (2mq\Delta V)^{1/2}/qB$$

Radius $\propto$ 1/curvature

- Time period ,

$$T = 2\pi m/qB$$

$$\text{Frequency} = 1/T = qB/2\pi m$$

- Angular velocity, $\omega = qB/m$

❖ Lorentz Force-

- when both electric and magnetic field are present then net force,

$$F_{\text{net}} = qE + qvB$$

If $F_{\text{net}} = 0$, then v = constant

$$v = E/B$$

- if $E = 0$ and $B = 0$ then, "v must be constant"

if $E \neq 0$ and $B \neq 0$ then "v can be constant or not"

if $E = 0$ and $B \neq 0$ then "v can be constant when θ b/w v and B is 0° or 180° "

❖ Cyclotron-

- used to accelerate heavy charge particles like 'α'-particle but not used to accelerate light particles like electrons.

- Cyclotron frequency,

$$v_c = qB/2\pi m$$

- Kinetic energy of ions,

$$K.E. = q^2 B^2 R^2 / 2m$$

- The operation of cyclotron is based on the fact that the time for one revolution of an ion is independent of its speed or radius of its orbit.

❖ Force on a current carrying conductor in external field-

$$F = ilB \sin\theta$$

θ = angle b/w l and B .

- Inside uniform field, F_{net} on closed loop is 'zero'.

❖ Torque on current loop-

$$\tau = NiAB \sin\theta$$

θ = angle b/w area(A) and magnetic field(B)

❖ Magnetic moment-

- $M = NiA$

$$\tau = MB \sin\theta$$

- Potential energy of current carrying loop-

$$U = -MB \cos\theta$$

- If $U \rightarrow$ minimum ($U = -MB$) then, stable equilibrium
- If $U \rightarrow$ maximum ($U = MB$) then, unstable equilibrium
- If $U \rightarrow$ zero ($U = 0$) then, neutral equilibrium

❖ Biot-savart Law-

➤ This law tells about magnetic field due to a “small current element”.

➤ Magnitude of magnetic field,

$$B = (\mu_0/4\pi) Idl \sin\theta /r^2$$

$$(\mu_0/4\pi = 10^{-7})$$

➤ This law is analogous to ‘coulomb’s law’.

➤ Direction of magnetic field can be given by “right hand thumb rule”.

❖ Magnetic field due to straight current carrying wire-

○ If wire is infinite,

$$B = \mu_0 i / 2\pi r$$

○ If wire is semi-infinite,

$$B = \mu_0 i / 4\pi r$$

❖ Magnetic field due to circular arc-

At centre , $B = \mu_0 i / 2R$ (complete circle)

At centre generally, $B = (\mu_0 i / 2R) (\theta/2\pi)$

➤ A wire of ‘fixed’ length is converted into a loop of single turn has magnetic field B_0 , now if turns become N then new magnetic field will be,

$$B' = N^2 B_0$$

❖ Magnetic field due to coil on its axis-

$$B = \mu_0 N i R^2 / 2(R^2 + x^2)^{3/2}$$

❖ Ampere’s Law-

➤ “this law states that line integral of magnetic field over any closed loop is μ_0 times the net current flowing inside the amperian loop”.

$$\oint B \cdot dl = \mu_0 I$$

➤ Ampere’s law can be apply for any shape of closed loop.

➤ It is used to find value of magnetic field for ‘symmetrical charge distribution’.

❖ Magnetic field due to cylindrical wire-

- Hollow wire, outside $\rightarrow B = \mu_0 i / 2\pi r$ ($r > R$)
 surface $\rightarrow B = \mu_0 i / 2\pi R$
 inside $\rightarrow B = 0$
- solid wire, outside and surface same as hollow wire
 inside $\rightarrow B = \mu_0 i r / 2\pi R^2$ ($r < R$)

❖ Force b/w two parallel wires-

$$F/l = \mu_0 I_1 I_2 / 2\pi r \quad (r = \text{distance b/w wires})$$

- If currents are in 'opposite' direction, then force will be 'repulsive'
If currents are in 'same' direction, then force will be 'attractive'

❖ Magnetic field due to Solenoid-

$$B = \mu_0 n i \quad (n = N/l)$$

Here, N = no. of turns

- Magnetic field at one of its end, $B = \mu_0 n i / 2$

❖ Magnetic field due to Toroid-

$$B = \mu_0 N i / 2\pi R \quad (R = a + b/2)$$

❖ magnetic field due to moving charge-

$$B = \mu_0 / 4\pi (qV \sin \theta / r^2)$$

❖ Gyro-magnetic ratio-

- Ratio of magnetic moment to angular momentum of a particle having charge q and mass m .

$$M/l = q/2m$$

- Gyro magnetic ratio is independent of shells 'n'

$$M = q/2m (nh/2\pi)$$

But magnetic moment, M is proportional to 'n'

- If $n=1$ then, $M = eh/4\pi m$
In this condition, M is known as “bohr magneton”

❖ Moving coil galvanometer-

- There is a soft iron core which not only makes the field radial but also increases strength of magnetic field.
- When current flows through the coil, a torque acts on it,

$$\tau = NIAB$$

- The deflection of galvanometer,

$$\phi = (NAB/k) I$$

$$\phi \propto I$$

here, k is torsional constant i.e. ‘restoring torque per unit twist’

- Current sensitivity ,

$$\phi/I = NAB/k$$

- Voltage sensitivity ,

$$\phi/V = (NAB/k) 1/R$$

Dr. Aman Tilak
MBBS, AIIMS Delhi

SUMMARY

1. The total force on a charge q moving with velocity $\mathbf{v}$ in the presence of magnetic and electric fields $\mathbf{B}$ and $\mathbf{E}$, respectively is called the *Lorentz force*. It is given by the expression:

$$\mathbf{F} = q (\mathbf{v} \times \mathbf{B} + \mathbf{E})$$

The magnetic force $q (\mathbf{v} \times \mathbf{B})$ is normal to $\mathbf{v}$ and work done by it is zero.

2. A straight conductor of length l and carrying a steady current I experiences a force $\mathbf{F}$ in a uniform external magnetic field $\mathbf{B}$,

$$\mathbf{F} = I \mathbf{l} \times \mathbf{B}$$

where $|\mathbf{l}| = l$ and the direction of $\mathbf{l}$ is given by the direction of the current.

3. In a uniform magnetic field $\mathbf{B}$, a charge q executes a circular orbit in a plane normal to $\mathbf{B}$. Its frequency of uniform circular motion is called the *cyclotron frequency* and is given by:

$$\nu_c = \frac{qB}{2\pi m}$$

This frequency is independent of the particle's speed and radius. This fact is exploited in a machine, the cyclotron, which is used to accelerate charged particles.

4. The *Biot-Savart law* asserts that the magnetic field $d\mathbf{B}$ due to an element $d\mathbf{l}$ carrying a steady current I at a point P at a distance r from the current element is:

$$d\mathbf{B} = \frac{\mu_0}{4\pi} I \frac{d\mathbf{l} \times \mathbf{r}}{r^3}$$

To obtain the total field at P, we must integrate this vector expression over the entire length of the conductor.

5. The magnitude of the magnetic field due to a circular coil of radius R carrying a current I at an axial distance x from the centre is

$$B = \frac{\mu_0 IR^2}{2(x^2 + R^2)^{3/2}}$$

At the centre this reduces to

$$B = \frac{\mu_0 I}{2R}$$

6. **Ampere's Circuital Law:** Let an open surface S be bounded by a loop C . Then the Ampere's law states that $\oint_C \mathbf{B} \cdot d\mathbf{l} = \mu_0 I$ where I refers to

the current passing through S . The sign of I is determined from the right-hand rule. We have discussed a simplified form of this law. If $\mathbf{B}$ is directed along the tangent to every point on the perimeter L of a closed curve and is constant in magnitude along perimeter then,

$$BL = \mu_0 I_e$$

where I_e is the net current enclosed by the closed circuit.

7. The magnitude of the magnetic field at a distance R from a long, straight wire carrying a current I is given by:

$$B = \frac{\mu_0 I}{2\pi R}$$

The field lines are circles concentric with the wire.

8. The magnitude of the field B inside a *long solenoid* carrying a current I is

$$B = \mu_0 nI$$

where n is the number of turns per unit length.

where N is the total number of turns and r is the average radius.

9. Parallel currents attract and anti-parallel currents repel.
10. A planar loop carrying a current I , having N closely wound turns, and an area A possesses a magnetic moment $\mathbf{m}$ where,

$$\mathbf{m} = N I \mathbf{A}$$

and the direction of $\mathbf{m}$ is given by the right-hand thumb rule : curl the palm of your right hand along the loop with the fingers pointing in the direction of the current. The thumb sticking out gives the direction of $\mathbf{m}$ (and $\mathbf{A}$)

When this loop is placed in a uniform magnetic field $\mathbf{B}$, the force $\mathbf{F}$ on it is: $F = 0$

And the torque on it is,

$$\tau = \mathbf{m} \times \mathbf{B}$$

In a moving coil galvanometer, this torque is balanced by a counter-torque due to a spring, yielding

$$k\phi = NIAB$$

where ϕ is the equilibrium deflection and k the torsion constant of the spring.

11. A moving coil galvanometer can be converted into a ammeter by introducing a shunt resistance r_s , of small value in parallel. It can be converted into a voltmeter by introducing a resistance of a large value in series.

POINTS TO PONDER

1. Electrostatic field lines originate at a positive charge and terminate at a negative charge or fade at infinity. Magnetic field lines always form closed loops.
2. The discussion in this Chapter holds only for steady currents which do not vary with time.
When currents vary with time Newton's third law is valid only if momentum carried by the electromagnetic field is taken into account.
3. Recall the expression for the Lorentz force,

$$\mathbf{F} = q (\mathbf{v} \times \mathbf{B} + \mathbf{E})$$

This velocity dependent force has occupied the attention of some of the greatest scientific thinkers. If one switches to a frame with instantaneous velocity $\mathbf{v}$, the magnetic part of the force vanishes. The motion of the charged particle is then explained by arguing that there exists an appropriate electric field in the new frame. We shall not discuss the details of this mechanism. However, we stress that the resolution of this paradox implies that electricity and magnetism are linked phenomena (*electromagnetism*) and that the Lorentz force expression *does not* imply a universal preferred frame of reference in nature.

4. Ampere's Circuital law is not independent of the Biot-Savart law. It can be derived from the Biot-Savart law. Its relationship to the Biot-Savart law is similar to the relationship between Gauss's law and Coulomb's law.

Important points from NCERT SUMMARY and Points To Ponder-

- A moving coil galvanometer can be converted into
 - 1) Ammeter → by introducing a small shunt resistance in parallel
 - 2) Voltmeter → by introducing a resistance of a large value in series
- Magnetic field lines always form closed loops.
- Ampere's circuital law is not independent of Biot Savart Law. It can be derived from the Biot Savart Law.